

What Are Information Systems?

An Overview

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Algorithm Basics and BW

What Is Technology?

- ▶ Application of knowledge to perform tasks
 - e.g., Wheel, Farming
- ▶ Embedded in business processes
 - e.g., The assembly line, The printing press
- ▶ Enables efficiency, consistency, and scale
 - e.g., Barcodes, double-entry bookkeeping

Not limited to gadgets, existed long before computers.

What Is Infrastructure?

- ▶ Shared foundation that supports technology
- ▶ Includes hardware, software, data, networks, people, rules, and governance
- ▶ Expensive, long-term, and difficult to change

Invisible when it works, critical when it fails.

Problem

*What happens when technology exists
but infrastructure doesn't support it?*

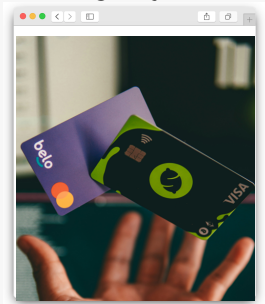
- ▶ Software rollout fails
- ▶ AI model gives bad outputs
- ▶ Implementation goes over budget

How did you pay for your last purchase?

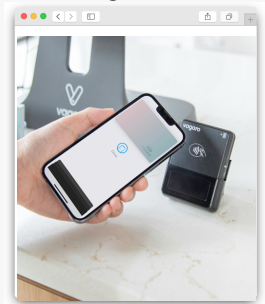
CASH



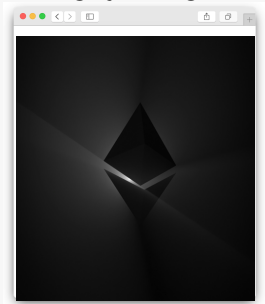
CARD



MOBILE



CRYPTO



WHAT TECHNOLOGY MADE THAT POSSIBLE?
WHAT IS THE INFRASTRUCTURE BEHIND EACH?

Technology in Finance - The Numbers

- ▶ **JPMorgan Chase:** \$18B technology budget (2025), or $\approx 10\%$ of revenue
- ▶ **Goldman Sachs:** 12,000 engineers (33% of workforce)
- ▶ **Morgan Stanley:** \$6.5B technology spending annually
- ▶ **Bank of America:** 19,000 developers (15% of workforce)

Technology Infrastructure in Finance

- ▶ Core banking systems - accounts, payments, fraud detection systems
- ▶ High-performance databases & data warehouses
- ▶ Regulatory & compliance platforms

Technology in Accounting - The Numbers

- ▶ **PwC**: AI audits 100% of transactions
- ▶ **Deloitte**: \$3B invested over five years in audit automation
- ▶ **EY**: $\approx 50\%$ of tasks automated
- ▶ **PwC, Deloitte, EY, KPMG**: 150,000+ technologists employed

Technology Infrastructure in Accounting

- ▶ Enterprise Resource Planning (ERP) systems
- ▶ Audit analytics & AI-based review tools
- ▶ Secure document management systems

Technology in Marketing - The Numbers

- ▶ **Amazon:** 35% of sales driven by recommendation algorithms
- ▶ **Coca-Cola:** 1B+ data points analyzed daily for marketing
- ▶ **Nike:** 40% of online revenue driven by data platforms
- ▶ **Unilever:** 100,000+ AI-optimized digital ad variants
- ▶ **P&G:** $\approx 20\%$ of marketing budget spent on technology

Technology Infrastructure in Marketing

- ▶ Customer Relationship Management (CRM) systems
- ▶ Customer profiling platforms and algorithms
- ▶ Digital advertising platforms

Technology in Manufacturing - The Numbers

- ▶ **Siemens:** €5B+ annual investment in automation and software
- ▶ **Tesla:** Software controls 80%+ of factory processes
- ▶ **General Electric:** 20% reduction in unplanned downtime via IoT

Technology Infrastructure in Manufacturing

- ▶ Industrial IoT sensors & control systems
- ▶ Supply chain & logistics systems
- ▶ Energy management systems

Technology in Management & Operations - The Numbers

- ▶ **McDonald's**: AI menu optimization increased sales by 5–10%
- ▶ **UPS**: Route optimization saves 10M gallons of fuel annually
- ▶ **Accenture**: \$1B annual investment in AI and cloud tools

Technology Infrastructure in Management & Operations

- ▶ Business intelligence & dashboard systems
- ▶ Enterprise data platforms
- ▶ KPI monitoring systems

Technology in Sports - The Numbers

- ▶ **Formula 1:** 300+ sensors per car, 1TB data per race
- ▶ **NBA:** 60 times a second, 29 points on every player
- ▶ **MLB:** Statcast records 100% of pitches and swings
- ▶ **NFL:** 30% reduction in injuries via analytics

Common Infrastructure Pattern Across All Industries

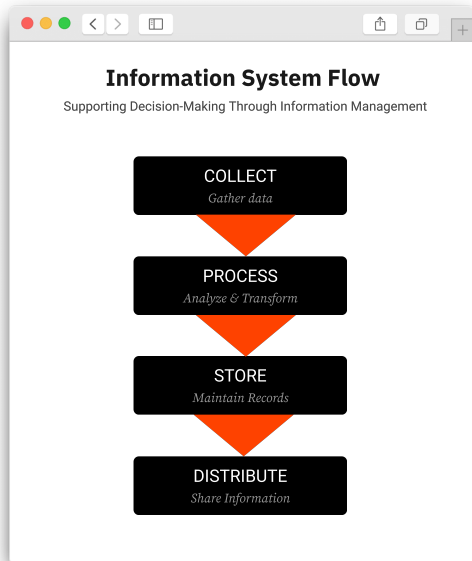
- ▶ Data collection systems
- ▶ Data storage & processing platforms
- ▶ Energy management systems
- ▶ Analytics & decision-support tools
- ▶ Governance, security, and controls
- ▶ Humans making final decisions

Why Study Information Systems?

- ▶ Organizations run on information
- ▶ Technology alone does not create value
- ▶ Advantage comes from how systems are used



What Is an Information System?

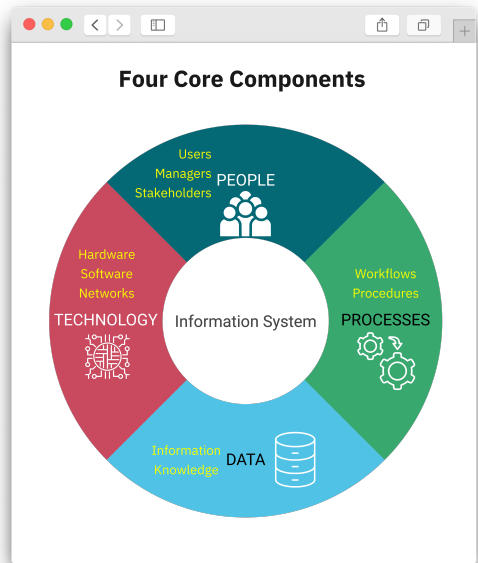


Definition

A coordinated set of components that **collect, process, store, and distribute information** to support decision-making.

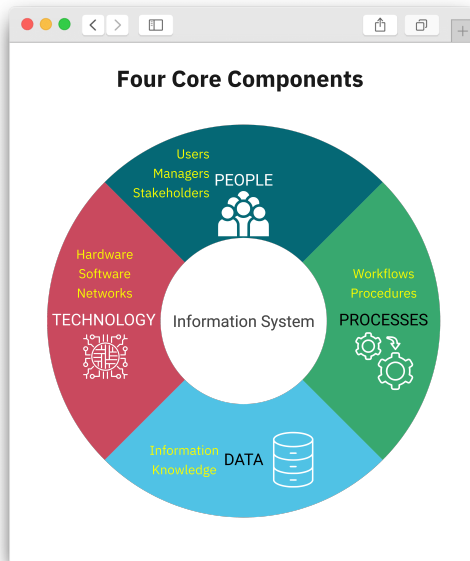
Core Components of an IS

1. People
2. Processes
3. Data
4. Technology



Core Components in Finance IS

1. People
 - ▶ Quant developers
 - ▶ Traders
2. Processes
 - ▶ Trade execution (microseconds)
 - ▶ Settlement ($T+1 = \$\$\$$)
3. Data
 - ▶ Market data feeds (\$500K+/month)
 - ▶ Alternative data (\$100K+/year)
4. Technology
 - ▶ Bloomberg ($\approx \$27\text{K}/\text{user}/\text{year}$)
 - ▶ Trading servers ($\approx \$500\text{K}$ each)
 - ▶ Data centers ($\approx \$1\text{B}+$)



Core Components in Accounting IS

1. People

- ▶ Accountants / auditors
- ▶ Compliance officers

2. Processes

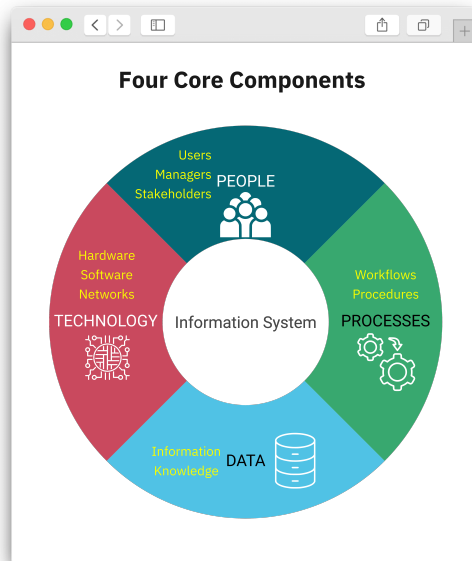
- ▶ Transaction recording
- ▶ Audit and reporting
- ▶ Financial consolidation

3. Data

- ▶ Ledger entries (\$millions transactions/year)
- ▶ Regulatory filings
- ▶ Historical audit data

4. Technology

- ▶ ERP systems ($\approx$ \$5M/year)
- ▶ AI audit platforms
- ▶ Document management systems



Core Components in Marketing IS

1. People

- ▶ Digital marketers
- ▶ Data analysts

2. Processes

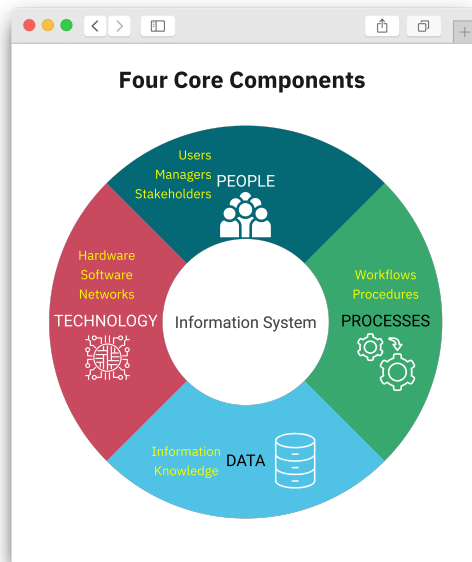
- ▶ Campaign management
- ▶ Customer segmentation
- ▶ Personalization workflows

3. Data

- ▶ Customer profiles
- ▶ Web traffic / clickstream data
- ▶ Social media analytics

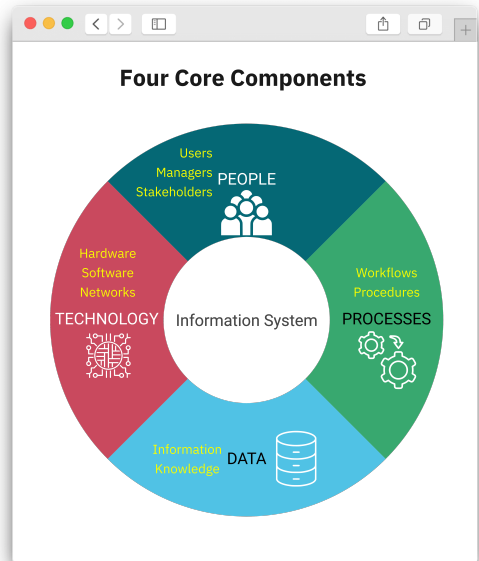
4. Technology

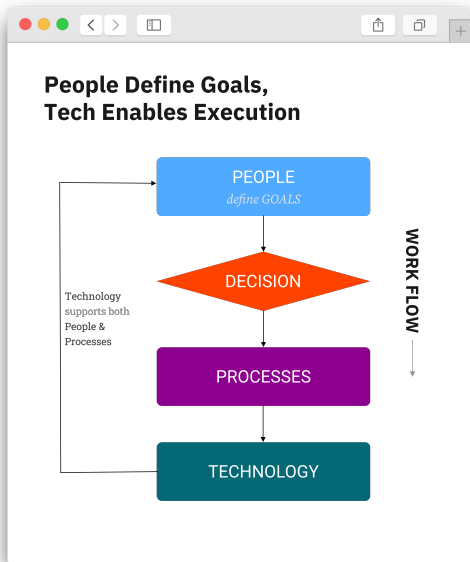
- ▶ CRM systems (\$50K–\$100K/year)
- ▶ Marketing automation tools
- ▶ AI recommendation engines



Core Components in Management & Operations IS

1. People
 - ▶ CEO & executives
2. Processes
 - ▶ Decision-making workflows
 - ▶ KPI tracking
 - ▶ Resource allocation
3. Data
 - ▶ Sales and performance metrics
 - ▶ Inventory levels
 - ▶ Process logs
4. Technology
 - ▶ BI dashboards (\$100K–\$500K/year)
 - ▶ Workflow automation platforms
 - ▶ Data warehouses





- ▶ People define goals and decisions
- ▶ Processes define how work flows
- ▶ Technology supports both

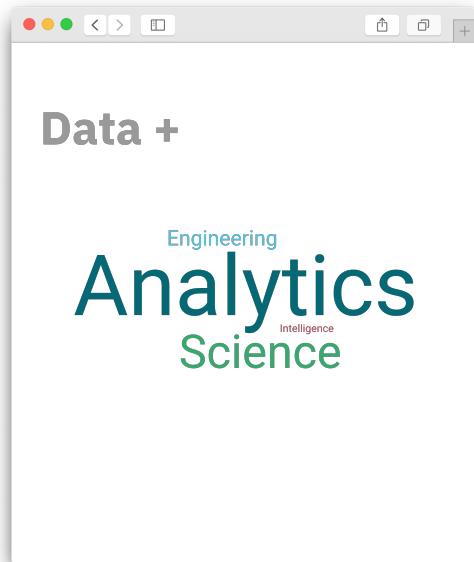
Process Example - Anatomy of a Stock Trade

Total visible time: 100 ms, **Behind the scenes:** 8+ systems, 20+ servers, 3 networks

Timeline of "Buy 100 AAPL @ Market":

T+0 ms: You tap "Buy" on phone
T+50 ms: Order reaches broker server (network latency)
T+52 ms: Broker validates (account balance, margin)
T+55 ms: Order routed to exchange (smart order routing)
T+56 ms: Exchange matching engine executes
T+57 ms: Confirmation back to broker
T+100 ms: You see "Order Filled" on screen
T+1 days: Settlement (cash/securities exchange)

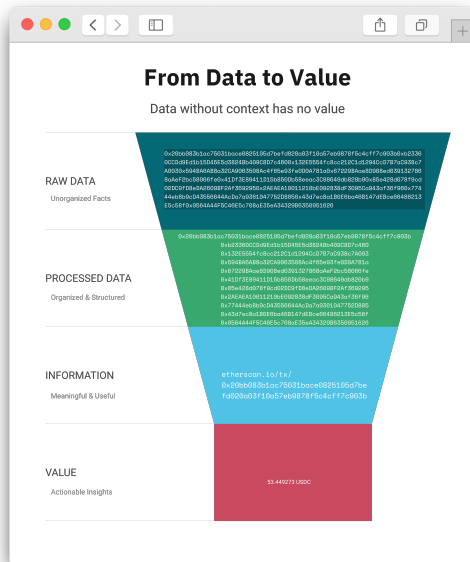
Terminologies



- ▶ **Analytics:** descriptive and diagnostic analysis
- ▶ **Science:** statistical, computational, and machine learning methods for actionable insights
- ▶ **Engineering:** designing, building, and maintaining the infrastructure that collects, stores, and processes data
- ▶ **Intelligence:** insights for business outcomes

Data and Technology

- ▶ Data are raw operational facts
- ▶ Technology processes data
- ▶ Data without context has no value



What is context? Data → Information → Knowledge - Finance Example

Data (raw):

AAPL: 259.50, 259.55, 260.48, 260.52, 259.60... (tick-by-tick)

Information (processed):

- AAPL VWAP today: \$260.06
- Volume: 46M shares
- Volatility: 1.2% (annualized: 19%)

Knowledge (contextual):

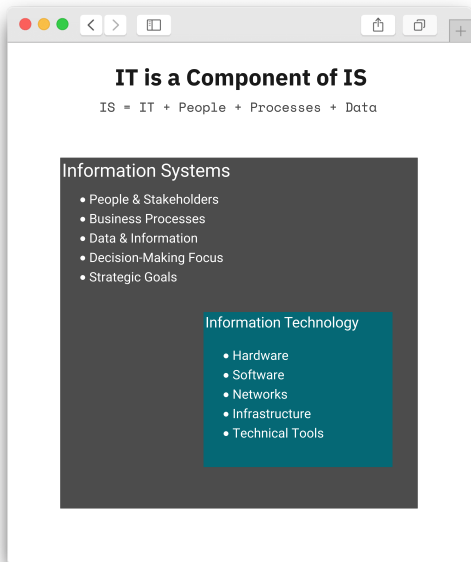
- AAPL typically rallies in Q4 (iPhone cycle)
- Positive correlation with NASDAQ (Beta = 1.2)

Wisdom (actionable):

- Consider AAPL long position for 2026 Q4

The Importance of Data Context in MIS

- ▶ Data alone is not information
- ▶ Context provides meaning
 - ▶ Who generated it? (source credibility)
 - ▶ When and where? (timeliness and relevance)
 - ▶ Why was it collected? (purpose and bias)
- ▶ Ignoring context can lead to poor decisions or costly errors



Information Technology (IT)

- ▶ Hardware
- ▶ Software
- ▶ Networks

Information Systems (IS)

- ▶ People + Processes + Data
- ▶ Technology as enabler
- ▶ Focus on decisions
- ▶ Context generation

How IS Creates Business Value

- ▶ Operational efficiency
- ▶ Better decision-making
- ▶ Strategic differentiation

